



ENME 441

PCB Design with KiCad

schematic → layout → fabrication

KiCad

KiCad is a free, open-source Electronic Design Automation (EDA) suite for designing schematics and printed circuit boards

1 Schematic Editor (Eeschema)

Capture your circuit as symbols connected by nets.

2 Footprint & symbol libraries

Thousands of built-in parts, plus custom and vendor libraries.

3 PCB Editor (Pcbnew)

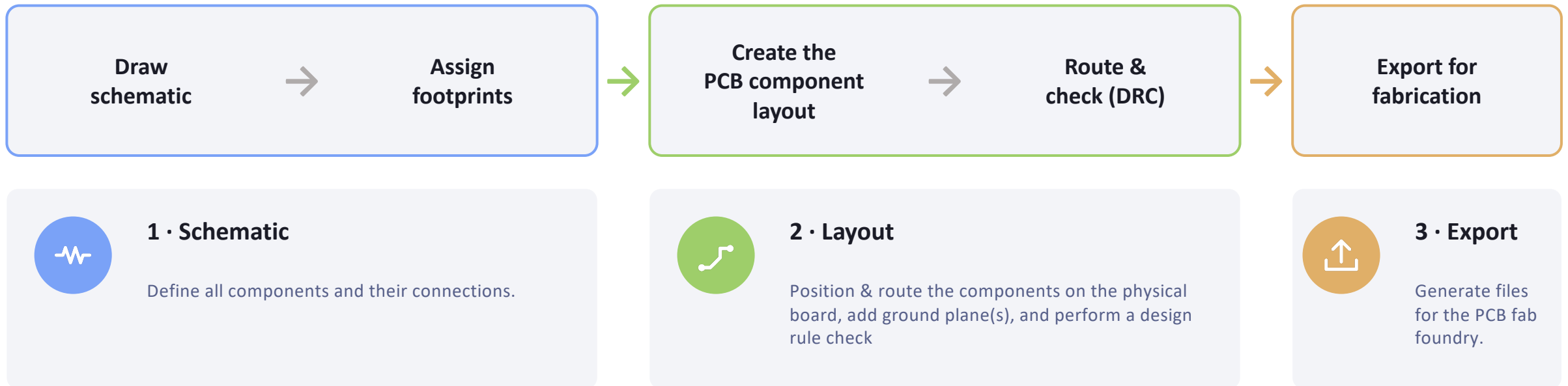
Place footprints and route copper on a physical board.

4 3D viewer & fabrication export

Preview the assembled board and output manufacturing files.



The KiCad Workflow



01

Schematic Capture

Define the circuit components and electrical connections



Schematic Editor Essentials

1

Place symbols

Drop components from the libraries into the schematic.

2

Wire the nets

Connect pins with wires; place junctions where wires connect.

3

Power & ground

Use power symbols (VCC, GND) instead of hand-wiring every rail. This will also make it easier to add ground zones during PCB layout

4

Labels & annotation

Name nets with labels; annotate to give every part a unique reference (R1, U2...).

Symbols vs. Footprints



Symbol

Defines the component pins and electrical connections. Allows creation of a netlist (connections list) for use in PCB layout.

Used during **schematic design**



Footprint

Defines the physical land pattern for the component (copper pad positions and overall part dimensions).

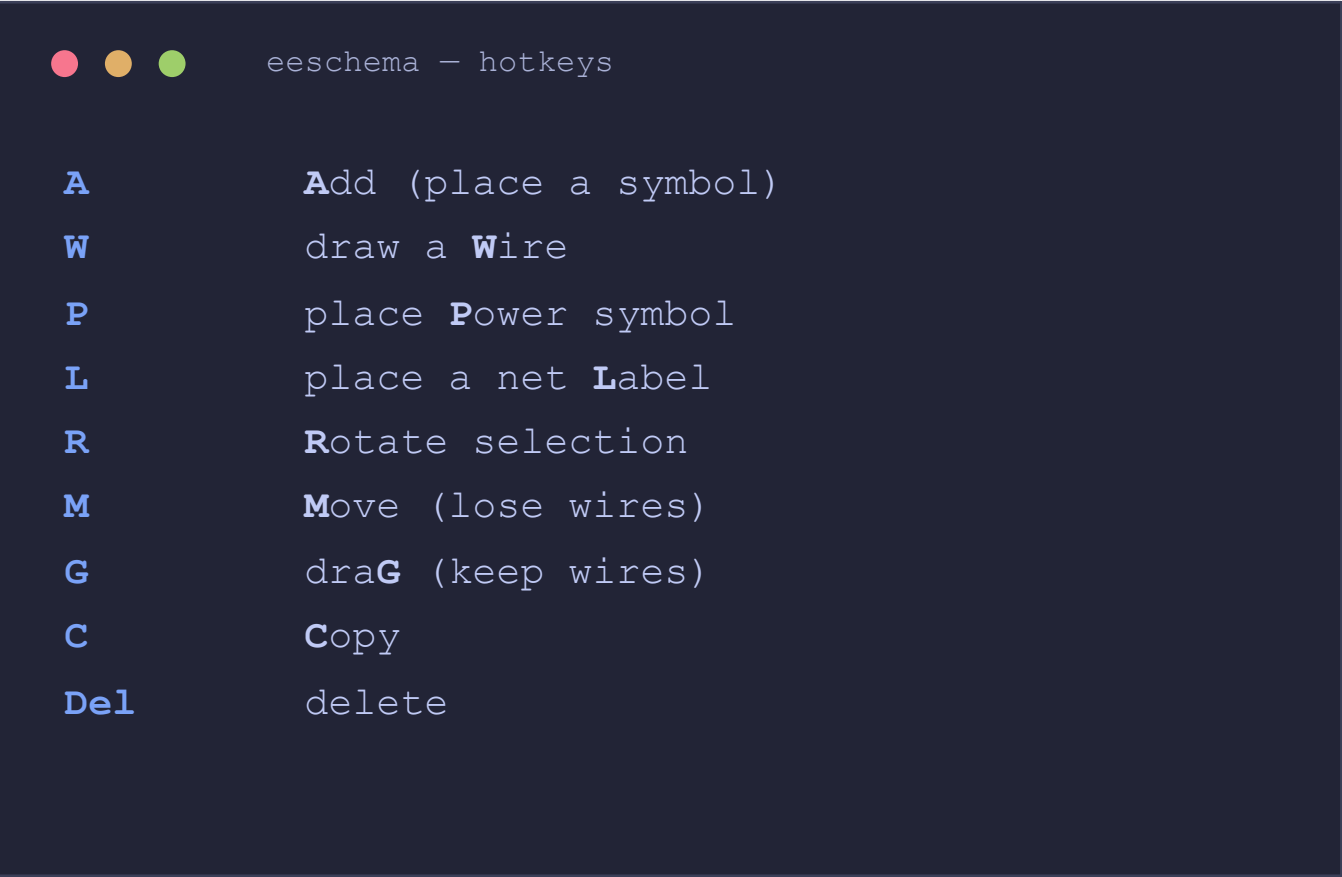
Used during **PCB layout**

Schematic **symbols** and PCB layout **footprints** are independent entities

A specific footprint must be assigned to each symbol to allow for PCB wiring

Allows for flexible many-to-one or one-to-many relationships

Work Fast: Key Hotkeys

A screenshot of the 'eeschema - hotkeys' window. The window has a dark background with light blue text. At the top left, there are three colored circles (red, orange, green) and the title 'eeschema - hotkeys'. Below this is a list of hotkeys. Each entry consists of a key name in all caps, followed by a description. The key names are color-coded: 'A' is blue, 'W' is orange, 'P' is green, 'L' is blue, 'R' is orange, 'M' is green, 'G' is blue, 'C' is orange, and 'Del' is blue. The descriptions are in a light blue monospace font.

A	A dd (place a symbol)
W	draw a W ire
P	place P ower symbol
L	place a net L abel
R	R otate selection
M	M ove (lose wires)
G	dra G (keep wires)
C	C opy
Del	delete



Hover, then press

KiCad hotkeys act on whatever is under the cursor
— no need to click on the component.



Same keys in both editors

Most hotkeys behave consistently in the schematic
and board editors.



Remap if desired

Preferences → Hotkeys lets you customize the full
list.

Electrical Rules Check (ERC)

Run ERC before moving to layout to catch connectivity mistakes early:

Inspect → Electrical Rules Checker

Select an error in the ERC dialog to highlight the error in the schematic



Unconnected pins

A pin left floating is often a forgotten wire or missing junction.



Conflicting outputs

Two output pins tied together



Power / ground flag

PWR_FLAG is a special symbol to tell the ERC where power comes from. Add one PWR_FLAG for each power symbol (GND, +5V, VCC, 3V3, etc.) on the single net wire where each power or ground is applied to the circuit.



Unassigned references

Symbols named R? / U? require annotation before generating a netlist.

02

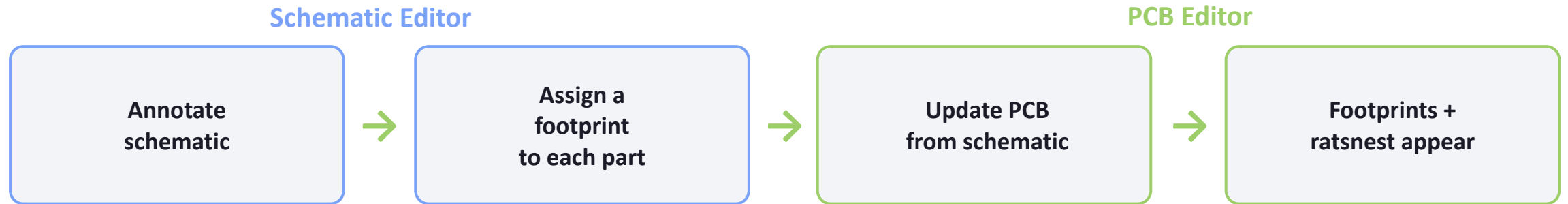
PCB Layout

Define physical positions and connections on the PCB



From Schematic to Board

KiCad pushes the finished schematic into the board editor for you.



Ratsnest — thin “rubber-band” lines showing the physical connections required for routing

Board Setup: Layers & Rules

1

Choose a layer count

Two copper layers (top + bottom) is typical for simple boards (more for dense designs)

2

Set the board outline

Draw the physical shape on the Edge.Cuts layer

3

Define design rules

Track width, clearance, and via sizes — match to fab house requirements.

4

Key layers

F / B	Front / Back sides (2-layer PCB)
Cu	copper (traces, planes)
mask	solder mask, defines regions where copper is exposed (automatic on device pads)
silkscreen	ink overlay (printed over solder mask)
edge.cuts	through-board cuts (overall board boundary, holes)

Placement & Routing

- **Place first, route second**

Arrange footprints logically — connectors at edges, decoupling caps near their ICs.

- **Follow the ratsnest**

Route the rubber-band connections into real copper tracks.

- **Manage layers with vias**

Drop a via to jump a track to another copper layer and avoid crossings.

- **Mind track width**

Wider tracks can carry more current and drop less voltage — size power nets accordingly.

- **Keep return paths short**

Route signals over continuous ground for clean, low-noise returns.

- **Keep signals away from WiFi antenna**

Avoid potential interference by adding a robust ground plane between signal traces and the location of the ESP32 WiFi antenna

Add ground zones

- Fill unused copper with a GND zone for shielding and easy return paths.
- For a 2-layer board create ground planes on both sides.
- Add vias to connect isolated islands.
- Zones don't auto-update — refill (B) before re-running DRC
- Route signals over continuous ground for clean, low-noise returns.

Design Rules Check

Pour ground planes, then let DRC verify the board is manufacturable.



Unrouted nets

Ratsnest lines mean connections aren't finished.



Clearance violations

Tracks or pads too close together — DRC flags anything below your rule.

03

Exporting for PCB fab



Fabrication Outputs

Once DRC passes, export the files the manufacturer needs.



Gerber files

One file per layer. Gerber is the universal PCB manufacturing format.



Drill files

Hole positions and sizes for vias and through-holes.



Bill of Materials

List of all components and quantities for assembly & ordering.



Placement (CPL)

Component positions for pick-and-place assembly.



3D model / STEP

Verify fit and export for mechanical CAD.



Order via fab house

Zip all Gerber files and upload to JLCPCB, OSH Park, etc.

Lab 10 – PCB Layout



We will perform this lab interactively. Submit your individual KiCad design for lab credit.

Create a schematic for a PCB to control a pair of 28BYJ-48 stepper motors through a SN74HC595 shift register and ULN2803 Darlington pair array IC, with an ESP32-C3 SuperMini used as the microcontroller. The SN74HC595 and ULN2803 will be soldered directly to the PCB. The ESP32-C3 will be connected with header pins to allow for easy swapping of boards. All components including the motors are driven using power from the ESP32 via a wired USB connection, so there is no need to include a separate power connector to the board.

The SN74HC595 comes in a 16-pin dual-in-line package (DIP) format, and the ULN2803 comes in an 18-pin DIP. Choose appropriate footprints (search for DIP-16 and DIP-18 in the footprint library).

Because the motor itself does not sit on the board, we need to include a connector. The 28BYJ-48 stepper motor comes with an XH-5P socket (a 5-pin, 2.54mm pitch JST XH-style female connector), so include a male 5-pin JST-XH connector in the schematic. The `conn_MxN_pin` component is a generic M row x N column male pin connector that can be used for this purpose, with the `JST_XH_B5B-XH-A_1x05_P2.50mm_Vertical` footprint applied to specify the actual pin layout in the PCB. The slight difference in pin pitch (2.54mm vs 2.50mm) will not affect performance.

Although the ESP32 will not be soldered to the board (since header pins will be used), we still need to include the ESP32 in the schematic and PCB layout to allow for proper wiring of all pins. Unfortunately, the ESP32-C3 SuperMini footprint is not provided in the standard KiCad library, so you will need to install this yourself. Download the ESP32-C3-SuperMini_KiCad package from Canvas and follow the instructions in the readme.md file (tip: open the markdown file using <https://mdview.io/> to see the formatting).